QUANTITATIVE RESEARCH DIVISION

CHATFIELD RESEARCH

Weekly Research Review & Division Expansion

Project progress, revised timeline and the next phase for Chatfield Quantitative Research.

Project Update
UK Vacancies Forecasting

Team Progress
Quants 1 – 3

Division Expansion
Recruitment & Term 2



Presented by **James Davies**

4 November 2026

Project Restructuring & Consolidation Week

Revised project timeline to support mid-term assessments and maintain research quality.

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1 Rationale

- **Mid-term assessments** for several analysts in the coming weeks.
- Need to **reduce workload** and allow time for consolidation and debugging.
- Maintain the **quality and consistency** of the final output.

2 Key Changes

- Introduce a **consolidation week** with no new tasks.
- Extend the project by **seven weeks**.
- Revised timeline and deliverables (see below).

3 Revised Timeline

Weeks 1 – 4	→	Week 5	→	Weeks 6 – 11	→	Week 12
Initial analysis and modeling		Consolidation (no new tasks)		Extended analysis, model refinement and evaluation		Final write-up, article and publication

4 Christmas Arrangements

- Project work **paused** over the Christmas period.
- Analysts to use this time for a **reset** and optional reading.
- Project will resume in Term 2 with a clear finalisation plan.

Key Takeaway

A short-term adjustment to reduce workload, consolidate progress and provide a clear, extended path to a high-quality final output.

Quant 1: Review of Last Week's Research

Data preparation and alignment for the UK labour market vacancies project.

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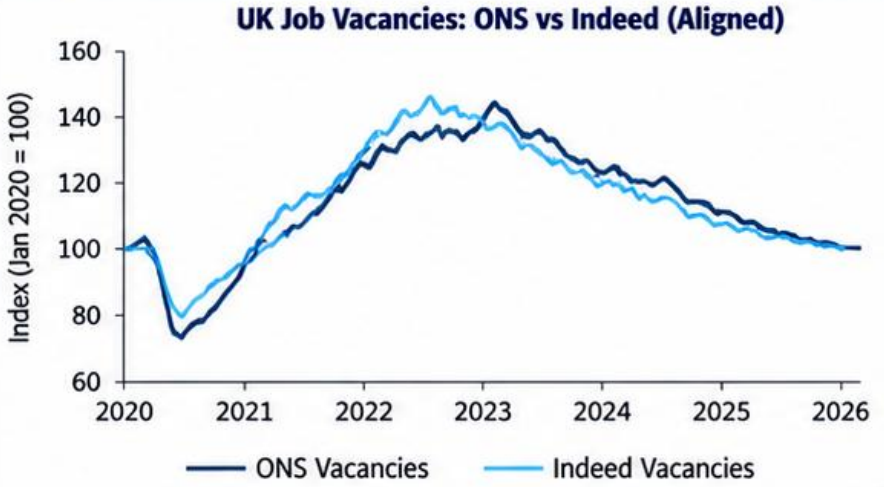
1 Objectives

- Collect and prepare ONS and Indeed vacancies data.
- Align series to a common time frame.
- Produce a clean dataset for forecasting.

2 Key Work Completed

- Downloaded and processed ONS and Indeed vacancies series.
- Aligned the data to **78** common reporting windows.
- Created a cleaned, documented dataset for model development.

3 Results and Observations



Key Findings

- Successfully aligned the two data sources over **78** reporting windows.
- Both series show consistent trends, with a peak in 2022–2023 and a gradual decline since.
- A clean, reproducible dataset has been created and documented.

Limitations / Outstanding Issues

- Differences in definitions and coverage across ONS and Indeed.
- Further feature construction required for modelling (e.g. lags and additional indicators).

Key Takeaway

A complete and aligned dataset has been produced, providing a solid foundation for forecasting and model development.

Quant 1: Next Week's Research Assignment

Validate the dataset, confirm variables and finalise documentation for modelling.

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1 Objectives

- **Validate** the aligned ONS–Indeed dataset constructed last week.
- Confirm the final set of **predictor variables** for the forecasting model.
- Ensure **data availability** and completeness for the full modelling period.

2 Key Tasks

- Run **consistency checks** (missing values, structural breaks and outliers).
- Define and justify the **predictor variables** (e.g. lags, additional labour market indicators).
- Confirm the final **data window** and update **documentation** (data dictionary and code comments).

3 Deliverables

Validated Dataset
Clean, aligned and ready for modelling.

Variable Specification
Final list of predictors with short justification.

Updated Code
Reproducible scripts with clear comments.

Documentation
Data dictionary and a brief method note.

4 Success Criteria

- A fully validated and well-documented dataset.
- A clear and justified set of predictor variables.
- All materials ready for integration into the modelling pipeline.

Key Takeaway

The focus this week is on validation and finalisation, ensuring a robust and well-documented dataset for forecasting.

Quant 2: Review of Last Week's Research

Forecasting the UK job vacancies series using a baseline time-series approach.

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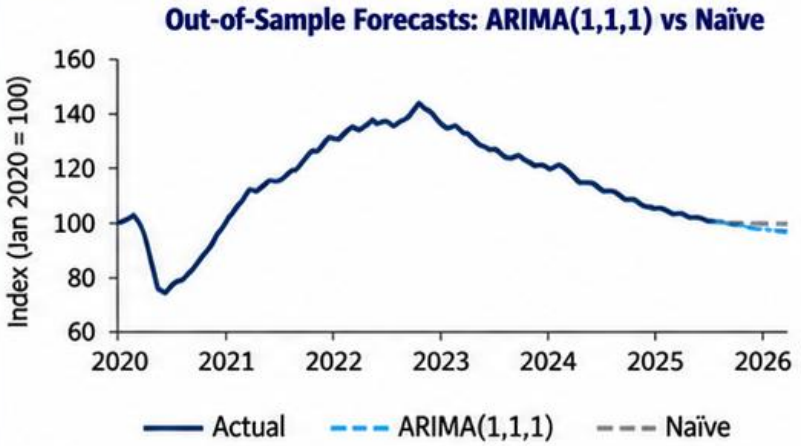
1 Objectives

- Build a simple forecasting pipeline for the vacancies series.
- Implement an **ARIMA(1,1,1)** model.
- Compare performance against a **naïve** (random walk) benchmark.

2 Methodology

- Used the aligned vacancies dataset (78 observations).
- Fitted an **ARIMA(1,1,1)** model (identified from initial ACF/PACF).
- Generated out-of-sample forecasts and evaluated using **MAE** and **RMSE**.

3 Results



Model	MAE	RMSE
ARIMA(1,1,1)	**3.42**	**4.67**
Naïve (Random Walk)	4.21	5.76

4 Key Takeaways

- **ARIMA(1,1,1)** outperforms the naïve benchmark on both MAE (3.42 vs 4.21) and **RMSE** (4.67 vs 5.76).
- The model captures the overall trend reasonably well.
- Provides a solid **baseline** for further model development and comparison.

5 Limitations

- A simple ARIMA specification with limited dynamics.
- Does not yet incorporate additional predictors (e.g. macroeconomic variables).
- Further diagnostics and model comparison are required.

Key Takeaway

A working forecasting pipeline has been developed, with ARIMA(1,1,1) providing a clear improvement over the naïve benchmark and a foundation for model extension next week.

Quant 2: Next Week's Research Assignment

Extend the forecasting analysis by incorporating ARIMA(1,1,2) and a formal model comparison.

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1 Objective

- Extend the current forecasting pipeline by adding an **ARIMA(1,1,2)** model.
- Compare **ARIMA(1,1,2)**, **ARIMA(1,1,1)** and the naïve benchmark to assess relative performance.

2 Key Tasks

- Implement and fit **ARIMA(1,1,2)** using the aligned vacancies dataset.
- Generate out-of-sample forecasts for all three models.
- Evaluate and compare performance using **MAE** and **RMSE**.

3 Analysis Framework

Model Estimation
Fit ARIMA(1,1,2) alongside ARIMA(1,1,1).

→

Forecasting
Generate out-of-sample forecasts.

→

Evaluation
Compute MAE and RMSE consistently.

→

Comparison
Assess relative performance and select the stronger specification.

4 Deliverables

- Updated code with ARIMA(1,1,2) and a standardised evaluation setup.
- A results table comparing MAE and RMSE for all three models.
- A short summary of findings and a recommendation for next steps.

Key Takeaway

The focus is on model extension and evaluation, providing a clear comparison between ARIMA(1,1,2), ARIMA(1,1,1) and the naïve benchmark.

Quant 3: Review of Last Week's Research

Time-series diagnostics and model selection for the UK job vacancies series.

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1 Objectives

- Analyse the statistical properties of the vacancies series.
- Use **stationarity tests** to inform appropriate model specification.
- Select and recommend an **ARIMA** model for forecasting.

2 Methodology

- Conducted Augmented Dickey–Fuller (**ADF**) and **KPSS** stationarity tests.
- Used **ACF/PACF** (post-differencing) to guide ARIMA model selection.
- Estimated candidate models and assessed **residual diagnostics** (ACF and Ljung–Box test).

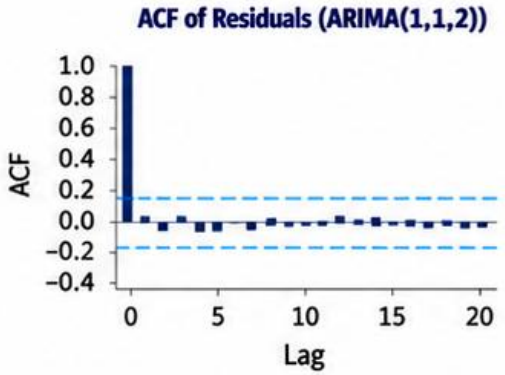
3 Key Results

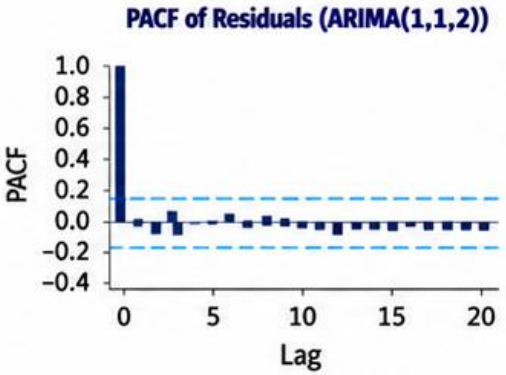
Stationarity Tests (Levels)

Test	Statistic	p-value	Conclusion
ADF	−1.24	0.66	Non-stationary
KPSS	1.18	< 0.01	Non-stationary

Model Comparison (Δy)

Model	MAE	RMSE	AIC
ARIMA(1,1,0)	3.71	5.18	878.4
ARIMA(1,1,1)	3.42	4.67	876.1
ARIMA(1,1,2)	**3.36**	**4.61**	**874.9**





4 Key Findings

- Both ADF and KPSS indicate the series is **non-stationary** at levels.
- Differencing **(d = 1)** achieves stationarity.
- Among the models tested, **ARIMA(1,1,2)** gives the lowest MAE, RMSE and AIC.
- Residual ACF/PACF show no strong remaining autocorrelation.
- **Recommendation:** proceed with ARIMA(1,1,2) for the forecasting pipeline.

Key Takeaway

The vacancies series is non-stationary, and ARIMA(1,1,2) is the recommended model based on information criteria, forecast accuracy and residual diagnostics.

Quant 3: Next Week's Research Assignment

Finalise the model recommendation, document diagnostics and prepare a methodological handover.

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1 Objectives

- Finalise and justify the model recommendation for forecasting.
- Document the full set of diagnostic results and their interpretation.
- Prepare a clear handover for integration into the final forecasting pipeline.

2 Key Tasks

- Confirm **ARIMA(1,1,2)** as the preferred specification (or provide evidence for an alternative).
- Summarise stationarity and residual diagnostics (ADF, KPSS, ACF/PACF).
- Write a concise methods note and update the project documentation.

3 Expected Outputs

Model Recommendation
Final model choice with **clear justification** (using AIC, MAE, RMSE and diagnostics).

Diagnostics Summary
Concise results and interpretation of ADF, KPSS, ACF and PACF.

Methods Note
A short, reproducible description of the analysis and model selection process.

Handover Materials
Clean code, annotated outputs and a summary for integration into the main pipeline.

4 Success Criteria

- A clearly justified and finalised model recommendation.
- A complete and well-documented set of diagnostic results.
- A concise methods note and handover ready for the team.

Key Takeaway

The focus is on finalising the model choice and producing clear, well-documented outputs to support integration into the wider forecasting analysis.

Introducing Quant 4: New Analyst & First Assignment

Welcoming the new Economics analyst and assigning their first research task.

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1 Welcome to the Team



- Introducing **Quant 4**, our new Economics analyst.
- Brings a strong **macroeconomic** and **policy perspective** to the team.
- Completes the core **four-analyst research group** for this project.

2 Role & Responsibilities



- Provide an **economic lens** on the UK labour market and vacancies.
- Coordinate with **Quants 1 – 3** to ensure a coherent, integrated analysis.
- Support the preparation of the final **article**, **slides** and **GitHub**.

3 First Research Assignment



Economic Context

Research key UK labour market developments over the sample period (e.g. policy changes, macro conditions).

→



Initial Analysis

Analyse simple relationships between vacancies and selected macroeconomic indicators (e.g. GDP, unemployment).

→



Output

A short, well-documented note with initial findings, key charts and suggestions for integration into the main project.

4 Collaboration & Support



- Work closely with all analysts and attend weekly research meetings.
- Use existing code and documentation as a starting point.
- Raise any data, code or methodology questions early.



Key Takeaway

Quant 4 strengthens the team's economics coverage, with an initial focus on UK labour market context and its link to vacancies, coordinated across the research group.

Next Steps: Priorities, Deadlines & Support

Focus for this week and key actions to keep the project on track.

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1 This Week's Priorities

- **Quant 2:** finalise ARIMA(1,1,2) and complete the model comparison table (MAE, RMSE, AIC).
- **Quant 3:** finish stationarity and residual diagnostics (ADF, KPSS, ACF/PACF) for the final model.
- **Quant 4:** complete initial economic analysis on the UK labour market and vacancies.
- **All:** update the GitHub repository and keep documentation consistent with the wider project structure.

2 Key Deadlines

Mon (3 Nov)	Code, results and updates from all analysts (18:00).
Tue (4 Nov)	Review and integrate results (Quant meeting, 18:00).
Wed (5 Nov)	Present findings and agree next steps (Quant session).
Next Week	Complete remaining analysis and finalise model specification for forecasting.

3 Support & Resources

- **GitHub:** project repository with code, data and documentation.
- **Past Materials:** Week 1 & 2 slides and working scripts for reference.
- **Data Sources:** ONS vacancies data and Indeed series (links in repo).
- **Questions:** raise on the WhatsApp group or at Tuesday's open session.

Key Takeaway

Deliver this week's analysis, keep the repository up to date, and build towards a final, integrated forecasting pipeline.

Key Takeaways & Next Steps

Summary of findings and the path forward for the project.

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1 Main Takeaways

ARIMA(1,1,2) as the **leading model:** best forecast accuracy and lowest information criteria among the models tested.

Non-stationary series: confirmed by ADF and KPSS, justifying the use of differenced ARIMA models.

Well-behaved residuals: ACF/PACF show no strong autocorrelation, supporting the model choice.

Clear path to extension: a solid foundation to incorporate additional models and economic indicators in the wider forecasting pipeline.

2 Immediate Next Steps (This Week)

- **Quant 2:** finalise ARIMA(1,1,2) implementation and model comparison.
- **Quant 3:** complete stationarity and residual diagnostics.
- **Quant 4:** deliver initial UK labour market analysis.
- **All:** update the GitHub repository and prepare for next week's integration.

3 Medium-Term Goals

- Integrate results across all analysts into a unified forecasting pipeline.
- Extend the model with additional approaches (e.g. VAR, Bayesian, or machine learning benchmarks).
- Incorporate further economic indicators (e.g. GDP, unemployment, policy data).
- Prepare the final outputs: article, presentation and GitHub release (Week 9–10).

Key Takeaway

We have a robust model and clear plan, and are on track to deliver a coherent, well-documented forecasting project for publication at the end of term.

Thank You & Discussion

Questions, feedback and suggestions for the next stage.

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1 Key Messages

- Completed a comprehensive time-series review of the UK jobvacancies series.
- The series is non-stationary, with ARIMA(1,1,2) recommended based on information criteria and diagnostics.
- Finalised the current analysis and assigned **Quant 4** to begin economic analysis this week.

2 Next Steps

This Week: Quant 2, 3 and 4 tasks and repository updates.

Next Week: integrate results and finalise the forecasting pipeline.

Medium Term: extend the model and prepare final outputs for publication.

3 Discussion

- Any questions on the results or model choice?
- Feedback on the planned next steps and division of work?
- Suggestions for additional analysis or extensions?

Key Takeaway

The project is on track, with a clear model, defined next steps and a strong foundation for a final forecasting analysis.